

CredentialChain

User Manual

How to issue, share and verify academic certificates – step by step, in simple words

What is CredentialChain?

CredentialChain is a website where a **college gives digital certificates** to students, and **anyone can check** in a few seconds whether a certificate is **real, fake or cancelled**.

Every certificate gets a secret code (a "fingerprint"). This code is saved in a **blockchain** – a special record book that nobody can change. If someone edits even one letter of a certificate, its code changes and the website says it is fake.

Who uses it?

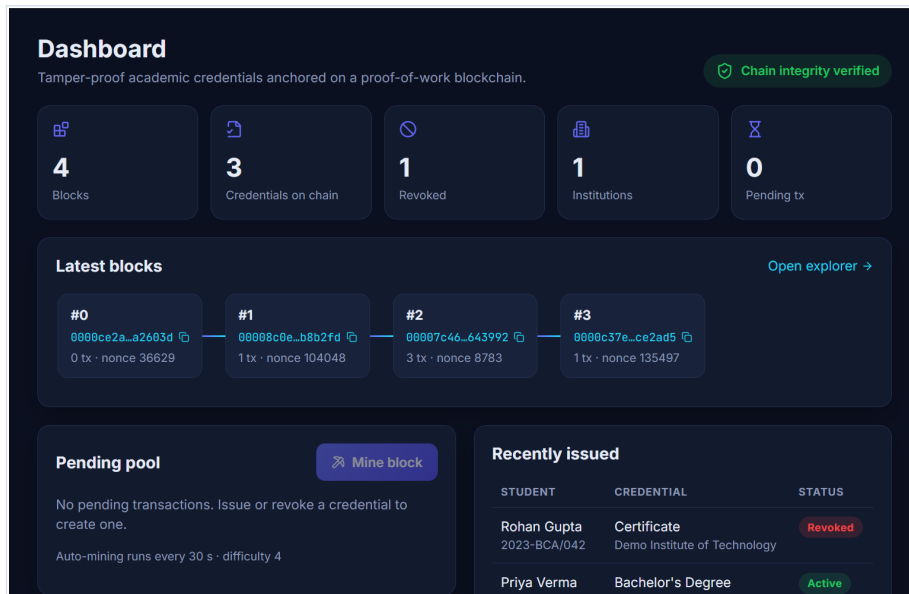
Who	What they do	Login needed?
Employer / anyone	Checks if a certificate is real	No
Student	Finds and shares their certificates	No
College staff	Makes, downloads and cancels certificates	Yes
Administrator	Adds new colleges	Yes

What is in this manual

Part	Read this if you are...
1. Opening the website	Everyone
2. Check a certificate	An employer or anyone checking a certificate
3. Find your certificates	A student
4. Make certificates	College staff
5. Add a new college	The administrator
6. Other pages	Anyone curious
7. Problems and answers	Anyone who gets stuck
8. Starting the software	The person who installs and runs it

1. Opening the website

- 1 Open **Google Chrome** (or any browser).
- 2 Type the website address you were given. On the computer running the software it is **localhost:5173**.
- 3 The **Dashboard** opens. Use the **menu on the left** to move between pages.



Picture 1: The Dashboard

The menu on the left

Menu item	What it is for
Dashboard	Summary: number of certificates, blocks, charts
Issuer Portal	College staff make and cancel certificates
Bulk Issue	College staff make many certificates from one Excel/CSV file
Verify	Anyone checks a certificate
Student Wallet	Students find their certificates by roll number
Chain Explorer	Look inside the blockchain
Institutions	List of colleges (admin adds new ones here)
Sign in	Login button at the bottom of the menu

Tip: The green badge "**Chain integrity verified**" at the top right means nobody has tampered with the records. If it ever turns red, contact the administrator.

2. Check a certificate (for employers)

You do **not** need an account. There are three ways to check. Use whichever is easiest.

Way A - Scan the QR code

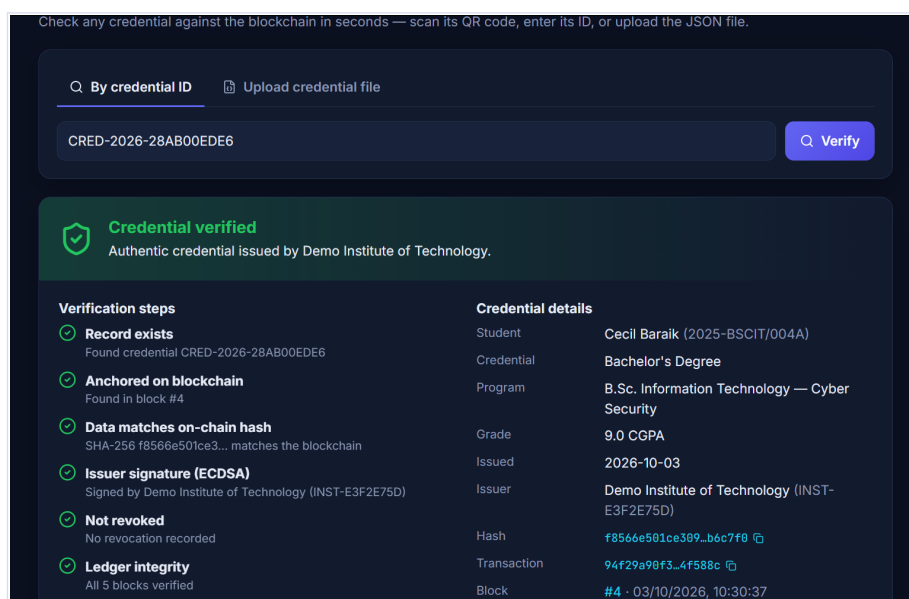
- 1 Open the camera on your phone.
- 2 Point it at the **QR code** printed on the certificate (bottom right).
- 3 Tap the link that appears. The result opens automatically.

Way B - Type the Credential ID

- 1 Click **Verify** in the left menu.
- 2 Type the **Credential ID** printed on the certificate, for example **CRED-2026-28AB00EDE6**.
- 3 Click the **Verify** button.

Way C - Upload the certificate file

- 1 Click **Verify**, then click the tab **Upload credential file**.
- 2 Click the dotted box and choose the **.json** file the student sent you.
- 3 The result appears straight away.



Picture 2: A real certificate: green banner and six green checks

Understanding the result

You see	Meaning	What to do
Green "Credential verified"	The certificate is real and unchanged	You can trust it
Yellow "Pending"	Real, but saved to the blockchain in a moment	Check again after 30 seconds
Red "Credential revoked"	Real, but the college cancelled it. The reason is shown	Do not accept it
Red "Verification failed"	The certificate was changed or faked	Do not accept it
"Credential not found"	No certificate with this ID exists	Check you typed the ID correctly

The **Verification steps** list shows each check with a green tick or a red cross, so you can see exactly what passed or failed.

The file is re-hashed and its signature checked against the issuer's on-chain public key.


Verification failed
 Verification failed — this credential cannot be trusted.

Verification steps

-  **Document integrity**
 Document content was edited: recomputed hash 2dd1407b089f... ≠ f8566e501ce3...
-  **Anchored on blockchain**
 Found in block #4
-  **Data matches on-chain hash**
 Data was altered: 2dd1407b089f... ≠ on-chain f8566e501ce3...
-  **Issuer signature (ECDSA)**
 Signed by Demo Institute of Technology (INST-E3F2E75D)
-  **Not revoked**
 No revocation recorded
-  **Ledger integrity**
 All 5 blocks verified

Credential details

Student	Cecil Baraik (2025-BSCIT/004A)
Credential	Bachelor's Degree
Program	B.Sc. Information Technology — Cyber Security
Grade	10.0 CGPA
Issued	2026-10-03
Issuer	Demo Institute of Technology (INST-E3F2E75D)
Hash	2dd1407b089f2a...f37812 🔗
Transaction	94f29a90f3...4f588c 🔗
Block	#4 · 03/10/2026, 10:30:37

Picture 3: A fake certificate: the grade was edited, so the checks turn red

3. Find your certificates (for students)

- 1 Click **Student Wallet** in the left menu.
- 2 Type your **roll number / student ID**, for example **2025-BSCIT/004A**, and click **Search**.
- 3 All your certificates appear. **Click one** to open it.

Sharing a certificate

On the certificate page you will find these buttons:

Button	What it gives you	Use it to
Download PDF	A printable certificate with a QR code	Print it or attach it to job applications
Download signed JSON	A small file with a digital seal	Send to employers – they upload it on Verify
Copy verify link	A web link to the result page	Paste in an email or message
Print	Prints the certificate page	Quick paper copy



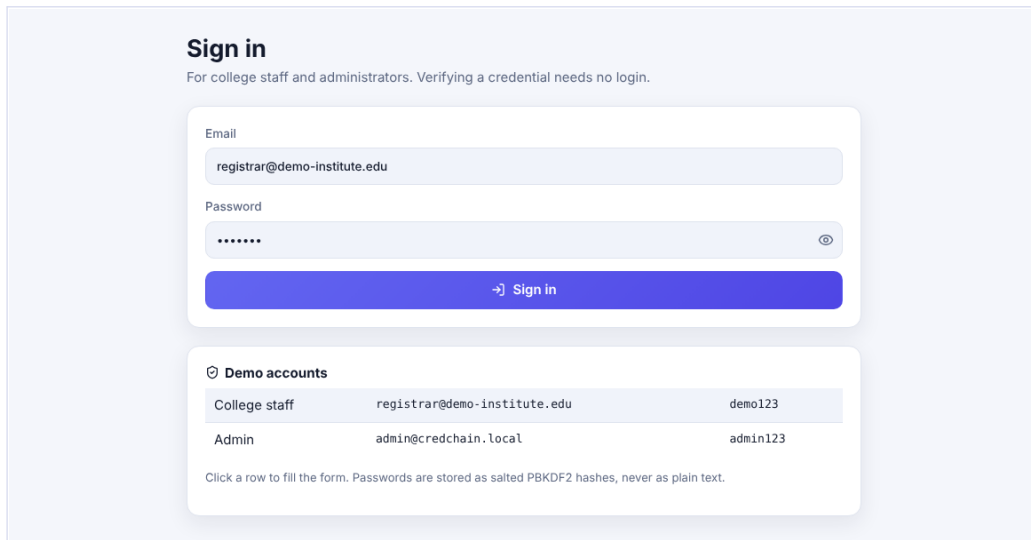
Picture 4: The PDF certificate (this example has been cancelled, so it shows a REVOKED stamp)

Important: Do **not** edit the JSON file. Any change – even a space in a grade – makes it fail verification.

4. Make certificates (for college staff)

4.1 Sign in

- 1 Click **Sign in** at the bottom of the left menu.
- 2 Enter the **email and password** the administrator gave you, then click **Sign in**.
- 3 You go straight to the **Issuer Portal**. Your name appears at the bottom of the menu.



The screenshot shows a 'Sign in' page with a heading 'Sign in' and a subtext 'For college staff and administrators. Verifying a credential needs no login.' Below this is a form with two input fields: 'Email' (containing 'registrar@demo-institute.edu') and 'Password' (containing six dots). A blue 'Sign in' button is below the password field. Underneath the form is a 'Demo accounts' section with a table of demo credentials.

Demo accounts		
College staff	registrar@demo-institute.edu	demo123
Admin	admin@credchain.local	admin123

Click a row to fill the form. Passwords are stored as salted PBKDF2 hashes, never as plain text.

Picture 5: The sign-in page

Important: After 5 wrong passwords the account is locked for 5 minutes. Wait and try again.

4.2 Make one certificate

- 1 In the **Issuer Portal**, choose the **Credential type** (Bachelor's Degree, Certificate, Diploma...).
- 2 Fill in **Student name**, **Student / roll ID**, **Program**, **Major** (optional) and **Grade**.
- 3 Choose the **Issue date**. Future dates are not allowed. Leave it empty for today.
- 4 Click **Sign & issue**. A preview of the certificate appears on the right with status **Pending**.

Student name

Student / roll ID

Program

B.Sc. Information Technology

Major / specialisation

Grade / CGPA

8.9 CGPA

Issue date

dd-mm-yyyy

Sign & issue

The record is hashed (SHA-256), signed with your ECDSA private key, and only the hash goes on-chain.

Cecil Baraik

(2025-BSCIT/004A) has successfully completed B.Sc. Information Technology with specialisation in Cyber Security

GRADE	ISSUED ON	CREDENTIAL ID	BLOCK
9.0 CGPA	2026-10-03	CRED-2026-28AB00EDE6	pending

SHA-256 FINGERPRINT

f8566e501ce3092a_b6c7f0

Scan to verify

Open credential page

Credentials issued by Demo Institute of Technology

ID	STUDENT	CREDENTIAL	BLOCK	STATUS
CRED-2026-28AB00EDE6	Cecil Baraik 2025-BSCIT/004A	Bachelor's Degree B.Sc. Information Technology	—	Pending Revoke

Picture 6: A certificate has just been issued – status Pending

4.3 Save it to the blockchain

A new certificate is **Pending** until it is saved into a block. This happens **automatically every 30 seconds**.

To do it immediately: go to the **Dashboard** and click **Mine block**. The status changes to **Active**.

Dashboard

Tamper-proof academic credentials anchored on a proof-of-work blockchain.

Chain integrity verified

5 Blocks

4 Credentials on chain

1 Revoked

1 Institutions

0 Pending tx

Latest blocks

Open explorer →

#0

#1

#2

#3

#4

Pending pool

Mine block

No pending transactions. Issue or revoke a credential to create one.

Auto-mining runs every 30 s · difficulty 4

Recently issued

STUDENT	CREDENTIAL	STATUS
Cecil Baraik 2025-BSCIT/004A	Bachelor's Degree Demo Institute of Technology	Active
Roban Gupta	Certificate	Revoked

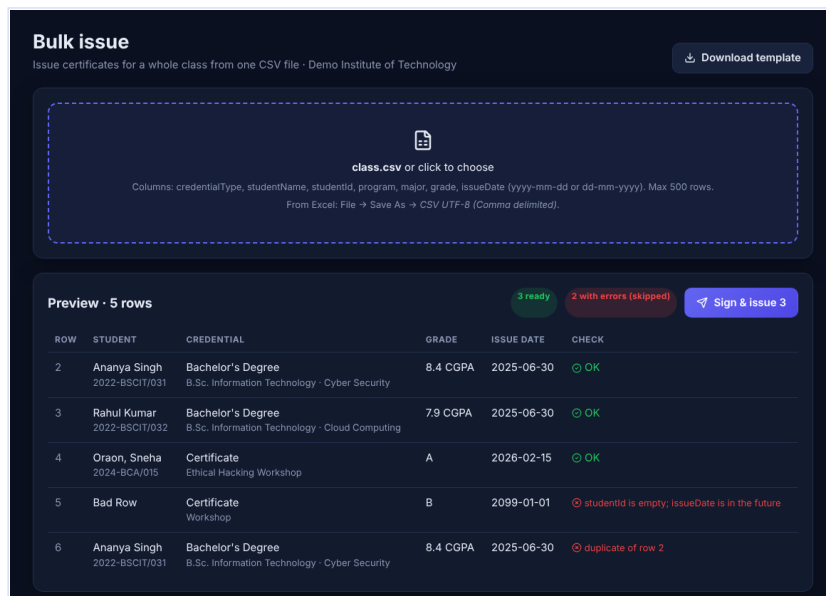
Picture 7: After "Mine block" a new block appears and the certificate becomes Active

4.4 Give the certificate to the student

- 1 In the table "**Credentials issued by ...**" at the bottom of the Issuer Portal, click the **Credential ID**.
- 2 Click **Download PDF** (and, if you like, **Download signed JSON**).
- 3 Send the files to the student, or tell them to find it in the **Student Wallet**.

4.5 Make many certificates at once (Bulk Issue)

- 1 Click **Bulk Issue** in the left menu, then **Download template**.
- 2 Open the template in **Excel**. Fill **one student per row**. Keep the first row (the column names).
- 3 In Excel choose **File → Save As → CSV UTF-8 (Comma delimited)**.
- 4 Drop the CSV file into the dotted box (or click it to choose the file).
- 5 Check the preview. Rows with problems are shown in **red** with the reason and will be **skipped**.
- 6 Click **Sign & issue**. When finished, click **Mine block now**.
- 7 Click **Download results CSV** to get every student's Credential ID and verify link.



Picture 8: Bulk Issue preview – 3 rows ready, 2 rows with errors

Column	Example	Required?
credentialType	Bachelor's Degree	Yes
studentName	Ananya Singh	Yes
studentId	2022-BSCIT/031	Yes

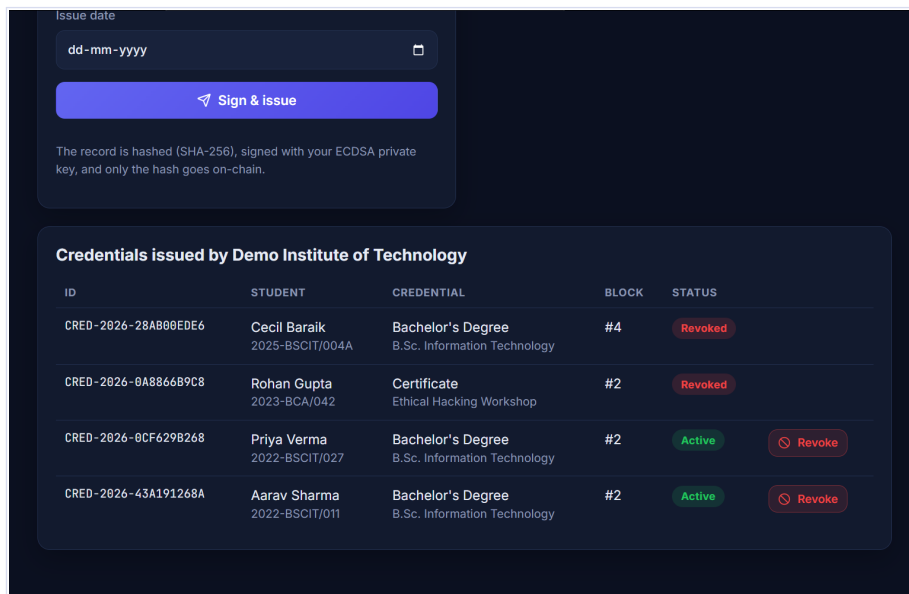
Column	Example	Required?
program	B.Sc. Information Technology	Yes
major	Cyber Security	No
grade	8.4 CGPA	Yes
issueDate	30-06-2025 or 2025-06-30	No (today if empty)

Tip: Up to 500 students per file. The same student with the same credential twice in one file is flagged as a duplicate.

4.6 Cancel (revoke) a certificate

Use this if a certificate was given by mistake. It is **not deleted** – it is marked as cancelled, and the reason is shown to anyone who checks it.

- 1 In the Issuer Portal, find the certificate in the table at the bottom.
- 2 Click the red **Revoke** button and type the **reason**, then click **OK**.
- 3 Mine a block (or wait 30 seconds). The status becomes **Revoked**.



Picture 9: Revoked certificates show a red "Revoked" label

Important: Revoking cannot be undone. If needed, issue a new, correct certificate instead.

4.7 Change your password

- 1 Click **your name** at the bottom of the left menu.
- 2 Under **Change password**, type your current password and the new one twice (at least 8 characters).
- 3 Click **Change password**.

4.8 Sign out

Click the **sign-out arrow** next to your name at the bottom of the menu. Always sign out on shared computers.

5. Add a new college (for the administrator)

- 1 Sign in with the **administrator** account.
- 2 Click **Institutions** in the left menu.
- 3 Fill in the college **Name**, a **Staff login email**, a **Staff login password** (at least 6 characters) and the website (optional).
- 4 Click **Generate keys & register**.
- 5 Give the **email and password** shown on screen to that college's staff. They can now sign in and issue certificates.

The system automatically creates a **digital key** for the college and saves it on the blockchain. Every certificate the college issues is "stamped" with this key.

Tip: An **API key** is also shown once. It is only for connecting other software; normal users can ignore it.

Attack lab (demonstration only)

In **Chain Explorer**, the administrator can use the **Attack lab** to show how the system catches cheating:

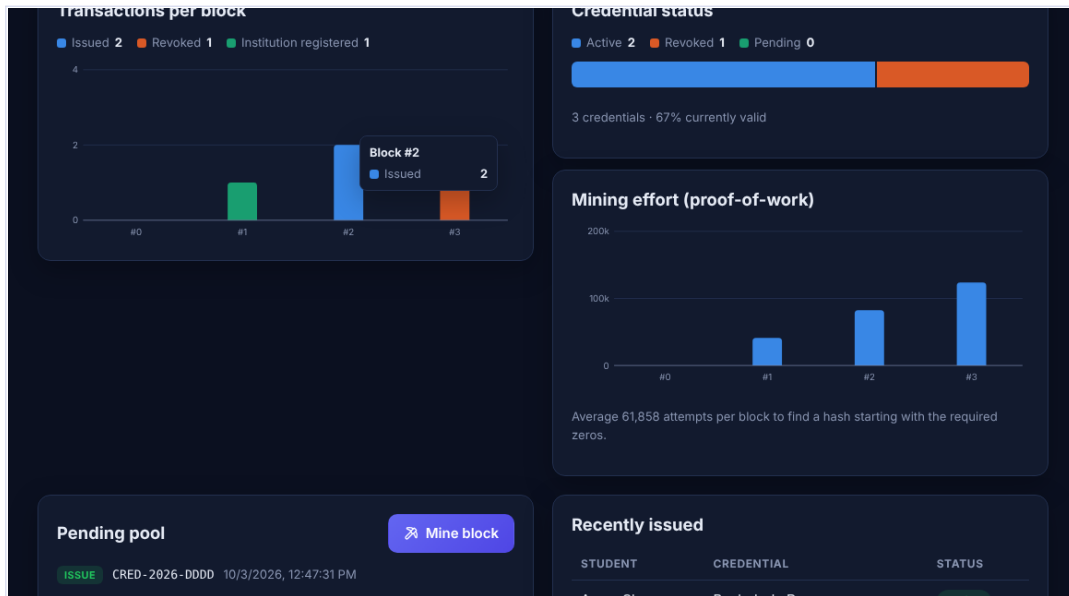
Button	What it simulates	Where to see it caught
Forge grade	Someone changes a grade in the database	Verify that certificate – it fails
Tamper block	Someone edits a saved block	Click "Validate entire chain" – the block turns red
Restore everything	Undoes both attacks	Chain is valid again

Important: Always click **Restore everything** after a demonstration, before closing the software.

6. Other pages

Dashboard

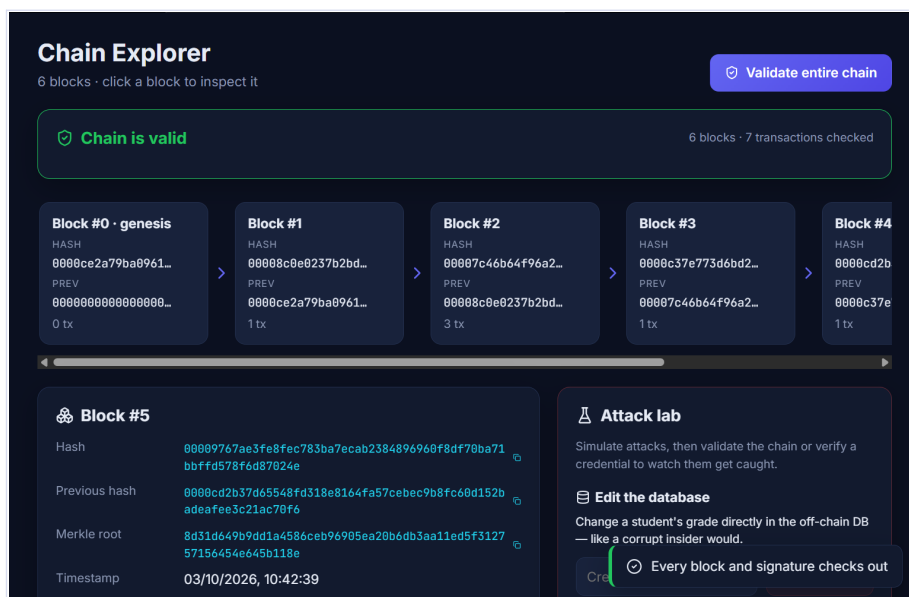
Shows totals (blocks, certificates, cancelled certificates, colleges), the latest blocks, three charts, the **Pending pool** (certificates waiting to be saved) and the most recent certificates. Hover over a chart bar to see its numbers.



Picture 10: Dashboard charts

Chain Explorer

Shows every block of the blockchain from left to right. Click a block to see its details and the records inside it. Click **Validate entire chain** to check that nothing has been changed.



Picture 11: Chain Explorer - "Chain is valid"

7. Problems and answers

Problem	Solution
The page says "Backend unreachable"	The software is not running. Ask the person who runs it to start it (see Part 8).
My certificate shows "Pending"	Wait 30 seconds, or ask staff to click "Mine block" on the Dashboard.
"Invalid email or password"	Check spelling and Caps Lock. After 5 wrong tries, wait 5 minutes.
"Credential not found"	Check the ID – it looks like CRED-2026-XXXXXXXXXX (zeros, not the letter O).
A real certificate shows "Verification failed"	The file was changed. Download a fresh copy from the website and try again.
QR code does not open on my phone	The website must be reachable from the phone (same Wi-Fi or a public link). Use the Credential ID instead.
Bulk upload says "Missing column(s)"	Use the template. Keep the first row with the column names, and save as CSV, not .xlsx.
"Issue date cannot be in the future"	Choose today or an earlier date.
I can't see the Issuer Portal form	Sign in with a college staff account – the admin account cannot issue certificates.

Quick questions

Can a certificate be deleted? No. It can only be cancelled (revoked), and the history stays.

Is my personal data on the blockchain? No. Only the secret code (fingerprint) is stored there.

Do employers need an account? No. Checking certificates is free and open to everyone.

8. Starting the software (for the person who runs it)

You need **Java 17 or newer**, **Node.js 18 or newer** and **IntelliJ IDEA** (or Maven) installed.

Start

- 1 Open the **backend** folder in IntelliJ and click the green **Run** button. Wait for "Started CredentialChainApplication".
- 2 Open a terminal in the **frontend** folder and type the commands below (the first one only the first time).
- 3 Open **localhost:5173** in the browser.

```
npm install
npm run dev
```

Stop

Click the red **Stop** button in IntelliJ, and press **Ctrl + C** in the terminal.

Default accounts (change these passwords!)

Role	Email	Password
Administrator	admin@credchain.local	admin123
College staff (demo college)	registrar@demo-institute.edu	demo123

Let phones scan QR codes

Same Wi-Fi: start the website with the command below and open the "Network" address it prints (e.g. <http://192.168.1.5:5173>).

```
npm run dev -- --host
```

From anywhere (free public link): install Cloudflare Tunnel once, then start it. Open the <https://....trycloudflare.com> link it prints.

```
winget install --id Cloudflare.cloudflared
cloudflared tunnel --url http://localhost:5173
```

Important: Open the website through the address phones can reach **before** making PDFs – the QR code uses the address you are on. Change the default passwords before sharing a public link.

Start fresh (delete all data)

Stop the software, delete the **data** folder inside the credential-chain folder, and start it again. Demo data is created automatically.